The Fog of Crisis: Asymmetric Linguistic Shocks in Executive Communication

Evidence from Interrupted Time Series and NLP Analysis of Earnings Calls (2012–2024)

Huseyin Mert Ezer

TU Dortmund

January 29, 2026

Research conducted for *Financial Econometrics WiSe25/26*

Agenda

1. **Introduction & Hypotheses**
   The "Fog of Crisis" and competing theories (Obfuscation vs. Rationality).
2. **Measuring the Shock**
   Quantifying the 2020 structural break using Interrupted Time Series (ITS).
3. **The Sectoral Divergence**
   Unsupervised detection of the "K-Shaped" linguistic split.
4. **The Counterfactual Analysis**
   Simulating a "No-Pandemic" world (ARIMA, Random Forest) to isolate the strategic residual.
5. **Conclusion**
   Synthesis: The paradox of "Resilient Complexity."

Introduction: The "Fog of Crisis" Hypothesis

1. The Research Question

- **The Phenomenon:** In March 2020, the global economy faced an unprecedented exogenous shock.
- **The Question:** Did this shock force executives into a uniform retreat behind complex language, or did it fracture communication styles along economic lines?
- **The Metric: Gunning Fog Index** (A proxy for linguistic complexity, measuring the years of formal education required to understand a text).

2. The Competing Hypotheses

H1: Strategic Obfuscation

- *Theory:* Managers use complexity to hide bad news.
- *Target:* **Vulnerable Sectors** (Physical Economy: Energy, Travel, Real Estate).
- *Prediction:* A massive spike in Fog due to "Defensive Vagueness."

H2: Rational Complexity

- *Theory:* Complexity reflects a genuinely complex new operating environment.
- *Target:* **Resilient Sectors** (Digital Economy: Tech, Finance, Staples).
- *Prediction:* A spike in Fog driven by the need to explain new paradigms (e.g., WFH, supply chains).

Methodology I: The Interrupted Time Series (ITS) Framework

1. **The Approach: Quantifying the "Shock"**
   - We treat the onset of COVID-19 (March 2020) as an exogenous "interruption" to the historical trajectory of communication.
   - **Goal:** To isolate the specific impact of the crisis from the **long-term underlying trend** (e.g., natural language evolution), ensuring we measure the shock rather than just the passage of time.

2. **The Econometric Model:**

$$Y_t = \beta_0 + \beta_1 T + \beta_2 \mathbf{D_t} + \beta_3 P_t + \epsilon_t$$

Variable Definitions:

- Y_t: Gunning Fog Index (Complexity) at time t
- T: **Time Trend** (The historical slope before the crisis)
- $\mathbf{D_t}$: **Crisis Dummy** (1 if Post-March 2020) $\rightarrow$ *Measures the Immediate Jump*
- P_t: Interaction Term ($T \times D_t$) $\rightarrow$ *Measures the Change in Trajectory*

Methodology I: Visualizing the Structural Break

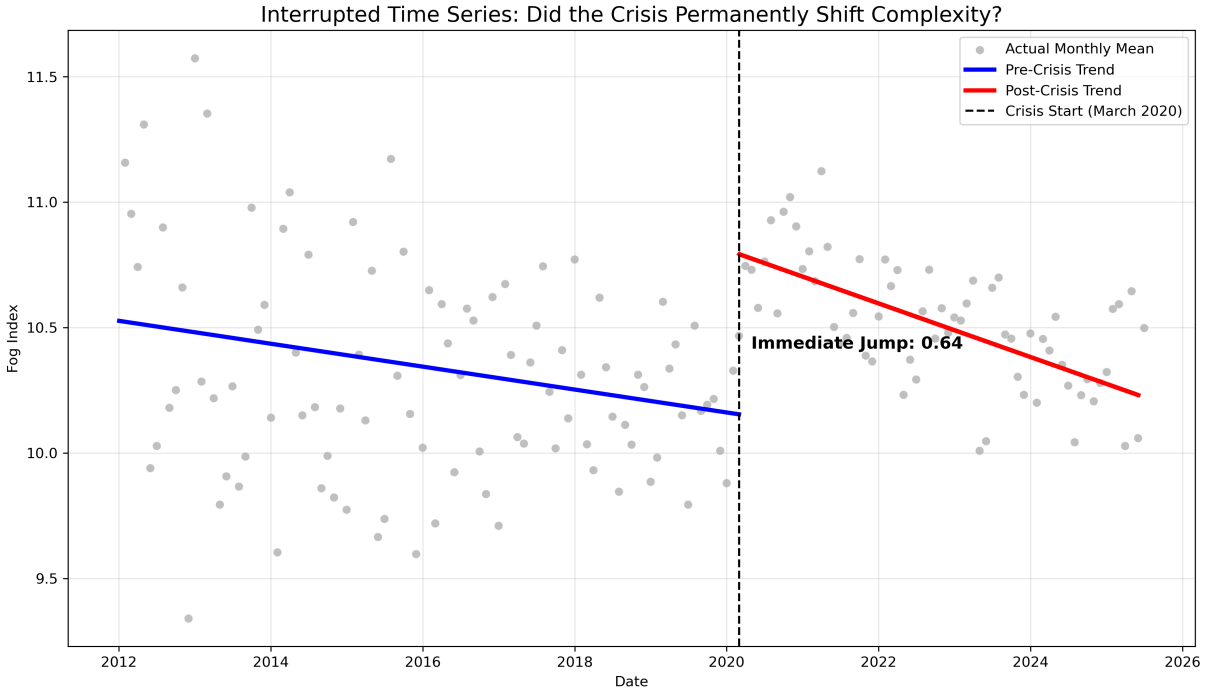

Figure 1: Interrupted Time Series Analysis of Gunning Fog Index (2012–2024)

Table 1: OLS Regression Results (Dependent Variable: Fog Index)

Variable	**Coefficient**	**Std. Error**	**t-Statistic**	**P-value**
Constant (β_0)	10.527	0.056	187.26	< 0.001
Secular Trend (T)	-0.0001	0.000	-4.54	< 0.001
Immediate Shock (β_2)	**+0.638**	**0.055**	**11.51**	< 0.001
Trajectory Change (β_3)	-0.0002	0.000	-3.76	< 0.001

Note: $N = 21,111$. $F = 45.87$. All coefficients are statistically significant at the 1% level.

Methodology II: Unsupervised Structural Break Detection

1. The Approach: "Letting the Data Decide"

- In the previous analysis, we *assumed* the break occurred in March 2020.
- **Objective:** Here, we remove all assumptions. We use a "Brute Force" search algorithm to mathematically identify the single point in time (t^*) that best splits the data into two distinct regimes.

2. The Algorithm (Piecewise Linear Regression)

1. Iterate through every possible quarter t from 2012 to Present.
2. Fit two separate linear models:
   - **Regime 1 (Pre-t):** $Y = \alpha_1 + \beta_1 X + \epsilon_1$
   - **Regime 2 (Post-t):** $Y = \alpha_2 + \beta_2 X + \epsilon_2$
3. Calculate the Sum of Squared Errors (SSE) for the combined model.
4. **Selection Rule:** The optimal break date t^* is the one that minimizes the Total Error:

$$t^* = \arg\min_t \left(\sum_{i=1}^{t} (y_i - \hat{y}_i)^2 + \sum_{j=t+1}^{n} (y_j - \hat{y}_j)^2 \right)$$

Methodology II: Visualizing the Regime Change

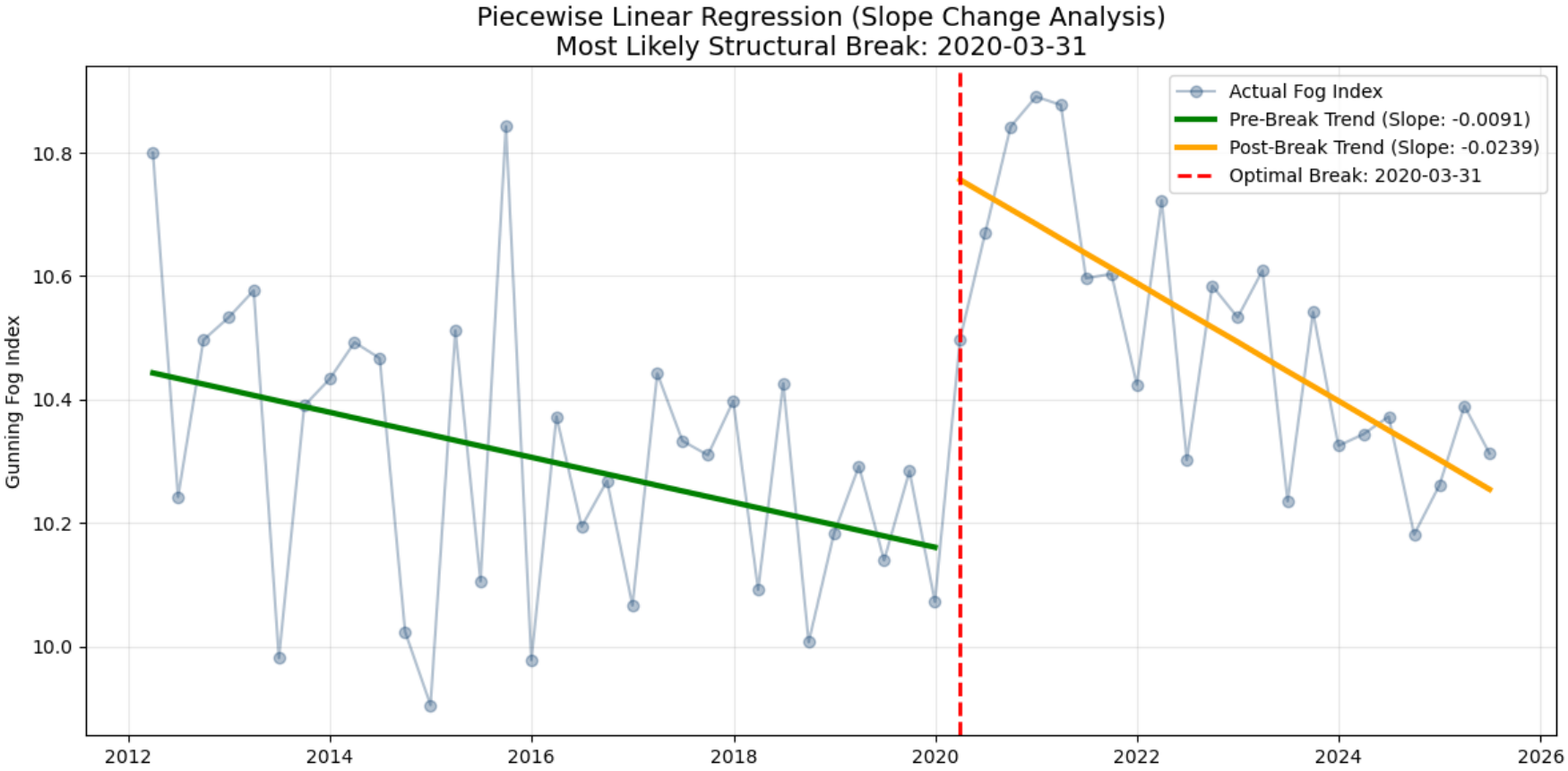

Figure 2: Optimal Structural Break Detection via Piecewise Linear Regression

Result: The algorithm autonomously identified **March 31, 2020** as the optimal structural break. Following the shock, the trend slope shifted from **-0.009** to **-0.024**, indicating an accelerated decline (reversion) rather than a permanent increase in complexity.

Sector Classification: The Ex-Ante Hypothesis

1. The Logic of Bifurcation

- The research suggests the pandemic was not a uniform shock but an "accelerant of bifurcation".
- We divide the GICS sectors into two theoretical cohorts based on their compatibility with social distancing and regulatory "harsh measures".

Cohort A: The Vulnerable (Physical)

- **Definition:** Sectors relying on physical presence, hit by demand destruction.
- **Sectors included:**
  - **Energy (10)**: Demand collapse.
  - **Real Estate (60)**: Rent crisis.
  - **Industrials (20)**: Airline grounding.
  - **Cons. Disc. (25)**: Service shutdown.

Cohort B: The Resilient (Digital)

- **Definition:** Sectors with digital capabilities or "essential" status.
- **Sectors included:**
  - **Info. Technology (45)**: WFH boom.
  - **Financials (40)**: Policy support.
  - **Comm. Svcs (50)**: Streaming/Gaming.
  - **Cons. Staples (30)**: Pantry loading.

Regression Results: The "Pandemic Break" Cluster

The Resulting Cluster (2020 Structural Breaks)

Table 2: Regression-Identified Structural Breaks (The "Pandemic Cluster")

Sector (Color from Slide 1)	Break Date (t^*)	Slope Change	Interpretation
Comm. Services (50)	2019-12-31	$-0.009 \to -0.038$	Complexity Shift (Streaming/Gaming)
Info. Technology (45)	2020-06-30	$+0.012 \to -0.036$	Reversal (Digital Infrastructure)
Financials (40)	2020-06-30	$-0.024 \to -0.036$	Acceleration (Credit Provisioning)

Observation: The "Pandemic Break" cluster consists *exclusively* of **Resilient (Blue)** sectors. The Vulnerable sectors experienced structural breaks in earlier years (e.g., 2015-2016), decoupled from the 2020 crisis.

Visualizing the Sector-Wise Structural Breaks

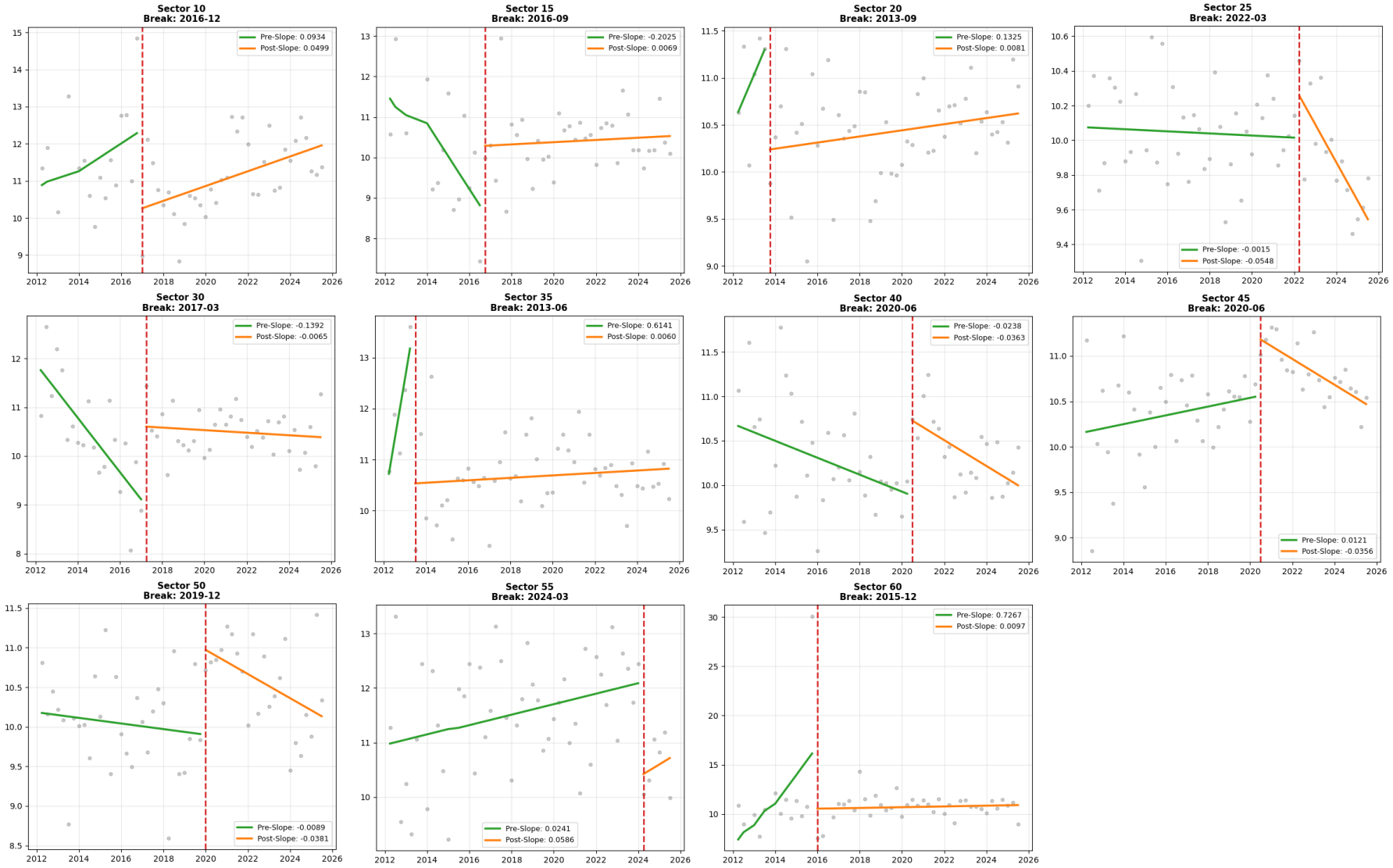

Figure 3: Comparative Analysis: Sector-Wise Structural Break Dates

Methodology III: The Counterfactual Simulation (ARIMA)

1. **The Objective: Generating "Business as Usual"**
   - **Goal:** To determine if the 2020 spike was *statistically abnormal* or merely random noise.
   - **Method:** We train an ARIMA model on pre-pandemic history (2012–2019) to simulate a theoretical world where the pandemic never occurred (the "Counterfactual").

2. **The Evaluation Test**
   - We project this forecast forward with a **95% Confidence Interval** (the orange tunnel).
   - **Decision Rule:**
     - **Inside Tunnel:** Normal Fluctuation.
     - **Breach Tunnel:** Statistically Significant Anomaly (The "Fog").

Methodology III: The Deviation from Normality

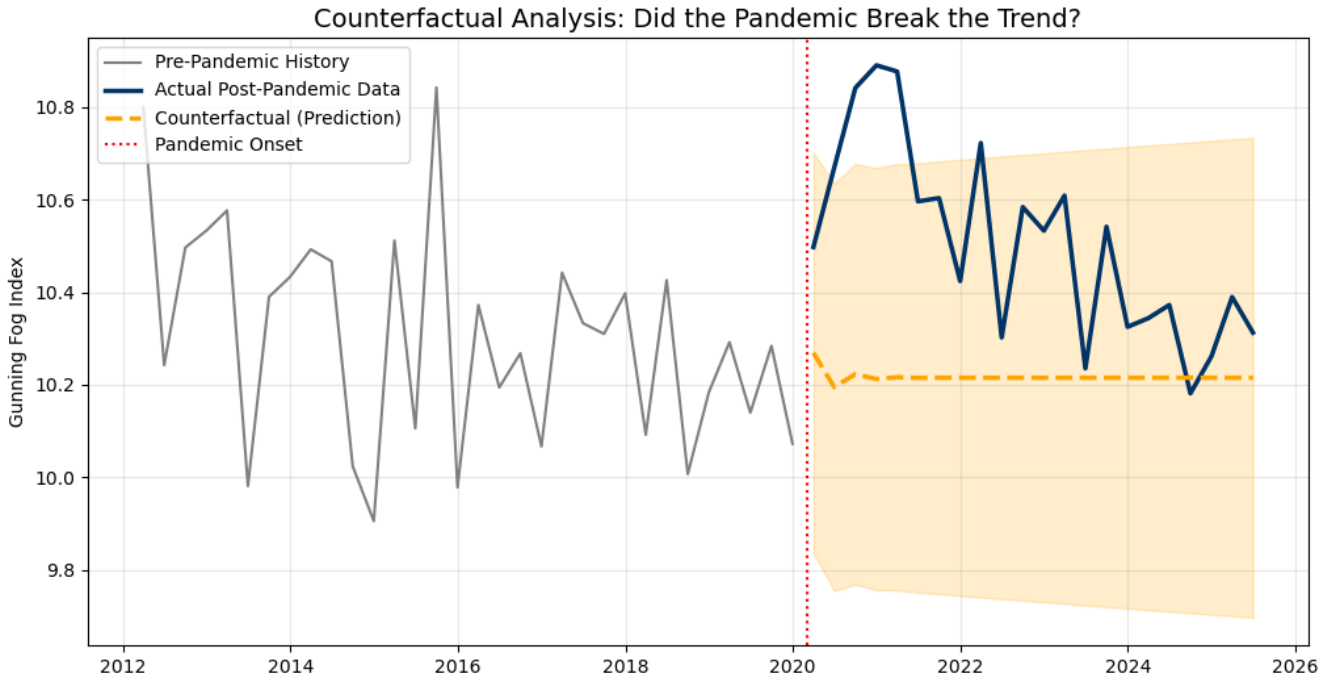

Figure 4: ARIMA Counterfactual Analysis: Actual vs. Predicted Fog Index

Analytical Interpretation:
The Actual Fog Index (Blue) briefly breaches the 95% Confidence Interval (Orange) in 2020, confirming a statistically significant exogenous shock. However, its return to the interval within approximately one year indicates that this "Fog of Crisis" was a **transient deviation**—driven by acute uncertainty—rather than a permanent structural shift in executive communication.

Drivers of Complexity: The Dominance of Sentiment

1. The "Obfuscation" Signal

- We analyzed which variables best predicted Fog Index from 2012–2019.
- **Result:** `speaker_finbert_negative` is the #1 predictor (17.6% importance), far outweighing market variables ($\approx 2\%$).
- **Implication:** Historically, complexity is a function of **Sentiment**, not just Market Volatility. When executives have "bad news," they reflexively use more complex language.

2. Top Feature Weights (Random Forest)

Table 3: Random Forest Feature Importance Ranking

Rank	Feature Name	Importance
1	**Negative Sentiment (FinBERT)**	**17.6%**
2	First Person Singular ("I/Me") Rate	15.2%
3	Singular Pronoun Ratio	9.6%
...	...	...
13	Prior Quarter Volatility	2.3%
14	Recent Volatility (30d)	2.2%

The "Unexplained" Crisis: Systematic Underestimation

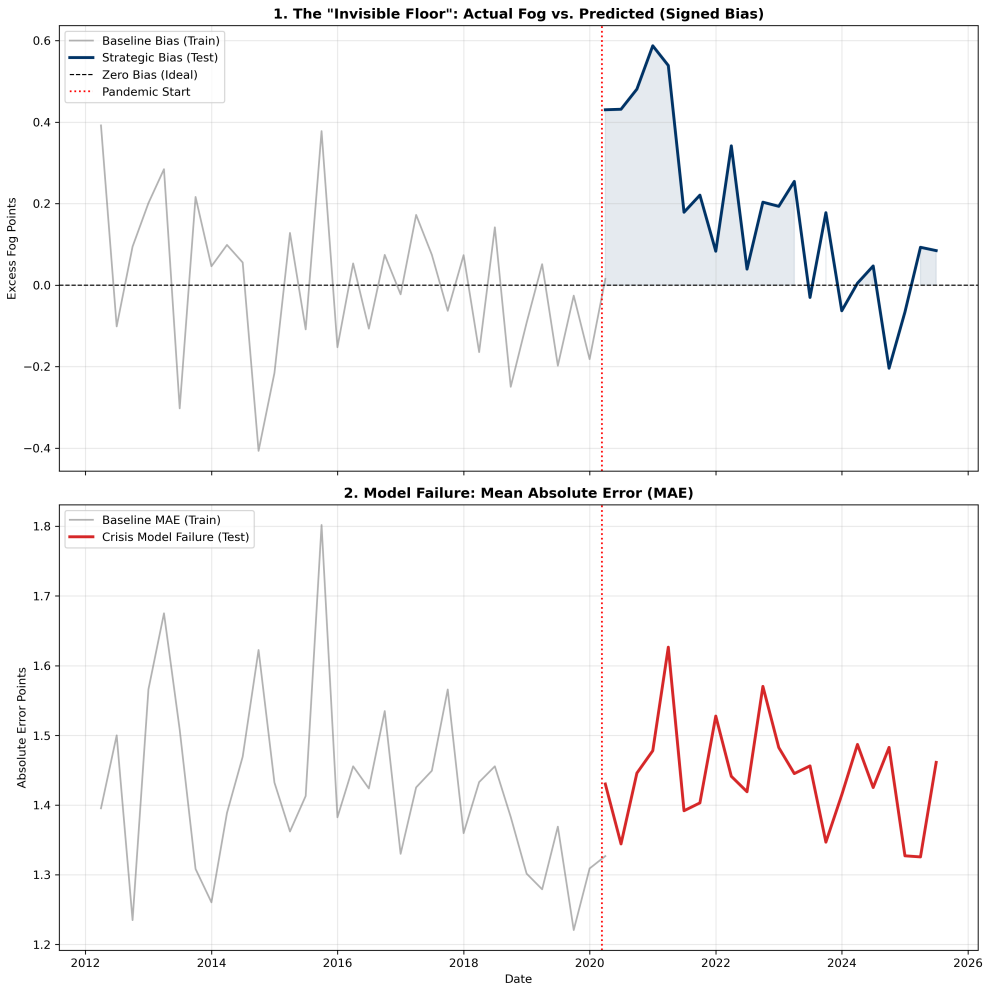

Figure 5: Residual Analysis: Signed Bias (Actual - Predicted)

Analytical Interpretation

Although the model generalized well (Train ≈ Test MAE), the *Signed Error* spiked positive in 2020. This indicates that while executives largely followed rational patterns, they systematically applied a **"Precautionary Markup"**: actual complexity consistently exceeded predictions, confirming a distinct layer of strategic fog on top of the rational response.

Conclusion: The Paradox of Resilient Complexity

1. **The Hypothesis Test (Sectoral Analysis)**

H1: Obfuscation (Rejected)

- **Evidence:** Vulnerable sectors (Energy, Real Estate) showed *no structural break* in 2020.
- **Verdict:** Struggling firms did *not* use the pandemic to hide behind complex language.

H2: Rationality (Supported)

- **Evidence:** The 2020 structural break was exclusive to Resilient sectors (Tech, Finance).
- **Verdict:** Complexity was driven by the *Digital Economy* articulating new operating paradigms.

2. **The Econometric Nuance: The "Fog Gap"**

- While the driver was rational (Digital Acceleration), the **Random Forest Residual** reveals a systematic underestimation, such that executives in resilient sectors used *more* complexity than sentiment or volatility warranted.

3. **Final Synthesis: "Precautionary Complexity"**

- The 2020 Fog was not about hiding *past failure* (Obfuscation), but about hedging against *future uncertainty*. The winners of the economy used complex language not to deceive, but to buy flexibility while rewriting the rules of the post-pandemic market.

References

- Papanikolaou, D., & Schmidt, L. D. (2022). Working remotely and the supply-side impact of Covid-19. *The Review of Asset Pricing Studies*, 12(1), 53–111.
- Lund, S., Madgavkar, A., Manyika, J., Smit, S., Ellingrud, K., Meaney, M., & Robinson, O. (2021). *The future of work after COVID-19*. McKinsey Global Institute.
- Global Industry Classification Standard. (2024, May 17). In *Wikipedia*. `https://en.wikipedia.org/wiki/Global_Industry_Classification_Standard`